

Task:- Design FS & Operation mode analysis.

Design the high-level software architecture for the payload subsystem. Your answer should include:

- Major software modules
- Responsibilities of each module
- Communication between modules
- External interfaces with the spacecraft bus
- External interfaces with the ground segment

Present the architecture using a block diagram accompanied by a brief explanation and, using the architecture designed above, define the operational software state machine for the payload. Include (considering entry, exit and operations):

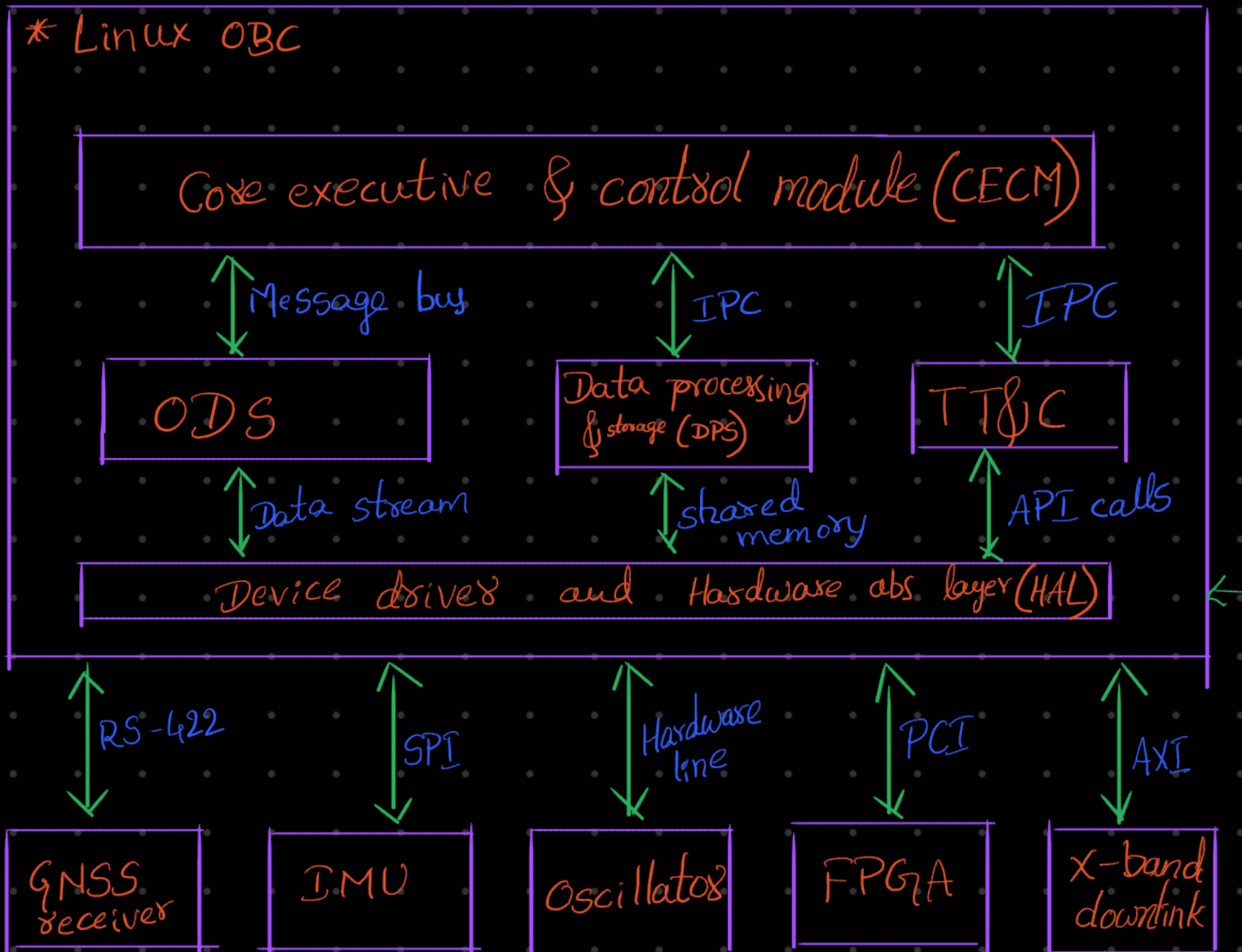
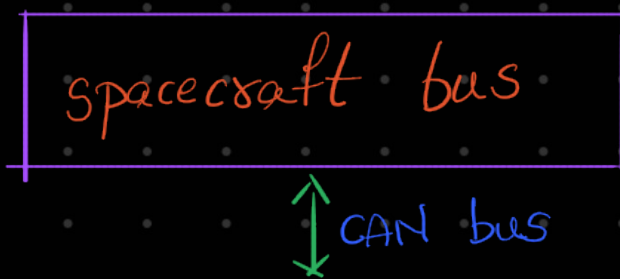
- BOOT
- SAFE
- COMMISSIONING
- NOMINAL
- DIAGNOSTIC

NASA cFs :- written entirely in C

Emerging alternative, "sat-ss" :- Written in rust & gaining traction.

↓
library called "embedded-hal" helps to write platform agnostic drivers.

Software architecture diagram.



CAN: Controller Area network

SPI: Serial Peripheral Interface

PCI: Peripheral Component Interconnect

AXI: Advanced extensible Interface

Rf-link

Ground Segment

S-band

[1] Core executive & Control module [CECM]:
This module governs critical tasks of the OBC. To name a few;
(a) System lifecycle management :- Runs the master state m/c that safely transitions through operational modes.

(b) FDIR :- [Fault detection, Isolation, and Recovery]
(.) protects the hardware by monitoring components health.

(c) Time synchronization :- Compares pulse from GNSS receivers & precise clock signals from oscillator & evaluates the accurate time.

[2] Orbit determination software [ODS]
Using advanced algorithms, computes the real-time position, velocity & time vectors.

[3] Data Processing & storage [DP&S]
Process the received data from sensors and save it to storage drives.

[4] Telemetry, Tracking & Command [TT&C] :-
(.) Builds the telemetry packets by gathering information from execution logs, ODS vectors, and diagnostic logs.

[5] Device drivers & Hardware Abstraction Layer [HAL]:
(.) Handles the low-level register signal by connecting them to API endpoint that include simple & standard commands.

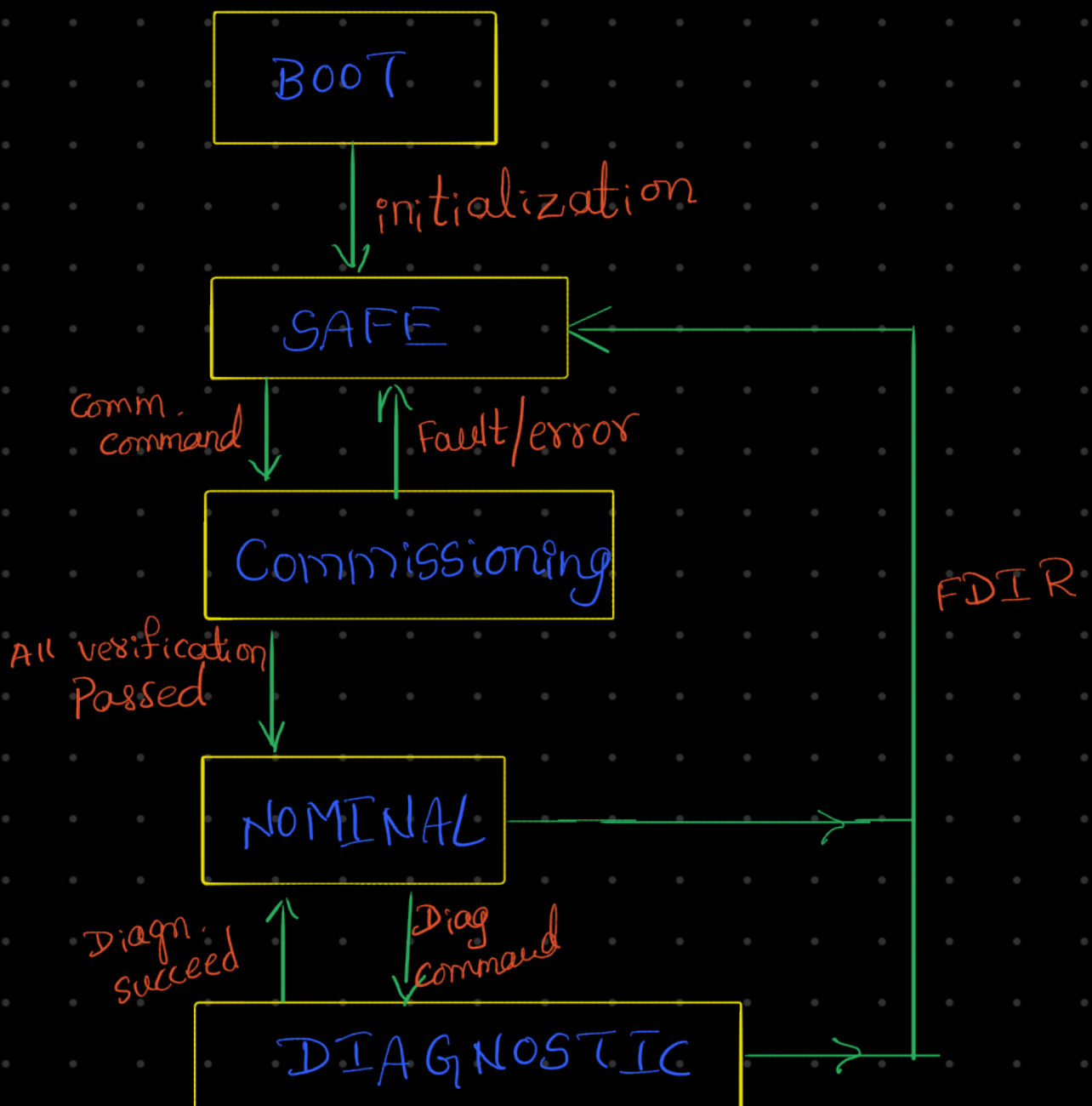
How does modules communicate?
[WIP]

External interfaces with the spacecraft bus.

Generally the satellite OBC interacts with spacecraft bus using CAN bus. All data packet exchange should follow CCSDS data protocol.

External interfaces with the ground stⁿ.
The ground control commands will be captured by the spacecraft transceiver & validate it before routing it to payload's TT&C module. For the downlink, all high-rate science packets will be relays back to ground station via X-band downlink transmitter. All data packet transfer should follow CCSDS file delivery protocol [CFDP].

state machine block diagram.



BOOT:- The entry to this state is when a power-on reset or hardware reset command is passed. During this state, the bootloader runs memory tests, boots the linux kernel & activates the hardware abstraction layers. Once the flight software is active, then the boot mode exits straight to safe mode.

SAFE:- As the name says, this mode is safe or power saver mode. safe mode is activated either by automation opⁿ from boot mode or from the fault detection mechanism. In this mode, operations are minimized to preserve power; sensors, FPGA and other non-essential components are powered off. The exit from this mode is possible with the verified command from ground and it enters into commissioning mode.

COMMISSIONING:- The entry point to this mode is strictly by safe mode. Operations like hardware bus link checks, FPGA component check and others are carried out. This mode exits to nominal mode iff all health checks are passed. If any one is failed, then it is immediately moved back to safe mode.

NOMINAL:- This mode is active either by successful commissioning checkout or from ground override from diagnostics. In this mode, all payload functionality are activated. This mode exits back to SAFE mode if any critical threshold are hit or transition into DIAGNOSTIC via direct command.

DIAGNOSTIC:— This mode is activated from the nominal mode for troubleshooting. operation like hardware loopbacks, memory cleanup, and also upload patched calibration matrices. The exit path returns to nominal mode upon ground command terminate the diagnostics or could end up in safe mode, if any test exception occurs.

↳ eg:- software update failed midway pushes the state to safe mode.

In conclusion, the architecture is designed to be simple. There could be modules which I may have overlooked. The communication terms is not clear to me. I tried to complete the block diagram to the extent I could understand. More to learn.

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